The Pauli group G_1 is relevant in quantum information theory, where

$$G_1 = \{\pm\mathbb{I}, \pm i\mathbb{I}, \pm\sigma_a, \pm i\sigma_a\},$$

and its presentation is

$$G_1 = \langle \sigma_1, \sigma_2, \sigma_3 | \sigma_a^2 = \mathbb{I}, \sigma_a\sigma_b = i\epsilon_{abc}\sigma_c \rangle$$

where the second relation holds for $a \neq b$.

4.5 Braid group

We can operate moves between pairs of strands, the most simple one is the Reidemeister move of the second type: undoing a passover.

Interchanging the strand i with $i+1$ (with i on the front) is denoted by σ_i. Its inverse is different (as we see with the Reidemeister move of the second type), and it is called σ_i^{-1}.

5 Lesson 5

5.1 Again, braid groups

The Reidemeister move of the first type consists simply in untangling a single strand with itself.

A generic braid can be written as a word

$$\sigma = \sigma_{j_1}^{i_1}\sigma_{j_2}^{i_2}\dots\sigma_{j_m}^{i_m},$$

with the generators $X_{n-1} = \{\sigma_1, \sigma_2, \dots, \sigma_{n-1}\}$.

Remark 5.1. The inverse of a word can be expressed as

$$\sigma^{-1} = \sigma_{j_m}^{-i_m}\dots\sigma_{j_2}^{-i_2}\sigma_{j_1}^{-i_1}.$$

To do something meaningful we need to have at least two strings. In general, a braid group with n strands is denoted as B_n.

With two strings the resulting group is already non-trivial

$$B_2 \cong (\mathbb{Z}, +).$$

Remark 5.2 (Commuting moves)**.** For all $n > 2$,

$$\sigma_i\sigma_j = \sigma_j\sigma_i$$

when $|i-j| \geq 2$. For instance when we have 4 strings

$$\sigma_1\sigma_3 = \sigma_3\sigma_1,$$

and that is intuitively true: the sets of strings on which they act are independent.

Remark 5.3. We can prove that (assuming $n > 2$)

$$\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}$$

for all i: the simplest way is just to perform the moves.

A similar relation exists in the field of integrable models: the Young-Baxter identities are based on the same type of structure.

In general, the presentation of a braid group is

$$B_n = \langle \sigma_1, \sigma_2, \dots, \sigma_{n-1} | \sigma_i \sigma_j = \sigma_j \sigma_i, \sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1} \rangle$$

for all $n > 2$.

5.2 Continuous groups

Definition 5.4 (Continuous group)**.** A group G is said to be continuous if its set contains an infinite and uncountable number of elements.

In general, continuous groups are not necessarily vector space (even if some of them are). To characterise their size we abuse the terminology, and define the *dimension* $\dim$ of G as the minimum number of independent real parameters (called *coordinates*) needed to parametrise all of its elements.

For example, the unitary group $U(1)$, which is based on the set of complex number that have modulo 1: it has dimension 1, because we can parametrise it with just one parameter.

We assume that when $g'' = g' \cdot g$, the coordinate of g'' depend smoothly on those of g and of g', this implies that the set of coordinates parametrising a group form a manifold, called the *group manifold*. If this is the case the group is said to be a *Lie group*.

For example, $(\mathbb{R}, +)$ is a continuous group that has dimension 1.[1] To get to higher dimensions we just need to multiply various copies of $\mathbb{R}$ where addition works on each component separately: $(\mathbb{R}^n, +)$ has dimension n.

If we take

$$[U(1)]^n = U(1) \times U(1) \times \ldots \times U(1) = \{(z_1, z_2, \dots, z_n) : |z_i| = 1, \forall i\},$$

we get a group of dimension n (because each component can be parametrised by a real number), which is isomorphic to a torus of dimension n.

We can also consider some matrix groups, like $\mathrm{Mat}(n \times m, \mathbb{R})$ with addition as the group product, which is a bit trivial and it has dimension $\dim = n \cdot m$. Other examples are Lie groups with matrix multiplication as the group product: all of these groups must only contain elements with $\det \neq 0$, otherwise the inverse matrix cannot be defined. There are the

1. general linear transformations (acting on $\mathbb{R}^n$) $GL(n, \mathbb{R})$, defined as[2][3]

$$\{A \in \mathrm{Mat}(n \times n, \mathbb{R}) : \det A \neq 0\},$$

2. special linear transformations $SL(n, \mathbb{R})$ defined as

$$\{A \in \mathrm{Mat}(n \times n, \mathbb{R}) : \det A = 1\}$$

[1] It is a distinct group from $U(1)$: the latter is periodic (among other things), thus they have a different global structure.

[2] The dimension of this group is still n^2, because the constraint is not an equality: this does not allow us to reduce the number of independent parameters.

[3] $GL(n, \mathbb{R})$ is a proper subgroup of $GL(n, \mathbb{C})$: we can just use the restriction of $\mathrm{Im} = 0$. The restriction $\mathrm{Re} = 0$ does not work, because the identity matrix would not be included.

which has dimension $n^2 - 1$,[4]

3. orthogonal group $O(n)$[5]
$$\left\{A \in \mathrm{Mat}(n \times n, \mathbb{R}) : A^T = A^{-1}\right\}$$
that has dimension $n(n-1)/2$,
4. unitary group $U(n)$
$$\left\{A \in \mathrm{Mat}(n \times n, \mathbb{C}) : A^{-1} = A^\dagger\right\}$$
that can be thought of as transformations of vectors in $\mathbb{C}^n$ that preserve their norm,[6][7]
5. special unitary group $SU(n)$
$$\{A \in U(n) : \det A = 1\},$$
which has dimension $n^2 - 1$ because now we are fixing $\det A = e^{i\alpha}$ (in general) to 1,
6. symplectic group $Sp(2n, \mathbb{R})$[8]

$$\left\{A \in \mathrm{Mat}(2n, \mathbb{R}), \Omega \text{ is a symplectic form} : A^T \Omega A = \Omega\right\}.$$

Remark 5.5. For $n = 1$, $Sp(2, \mathbb{R}) = SL(2, \mathbb{R})$.

So far we have talked about families of matrix Lie groups. Now we mention some "exceptional" groups: G_2, F_4, E_6, E_7, E_8: these groups were discovered when the Lie algebras of all continuous groups were classified. They have some applications in physics, particularly in string theory.

5.3 Actions of groups on a set

Definition 5.6 (Left action)**.** Considering a group G and a (non-empty) set X, we define a left action L of G on X as the group homomorphism

$$\begin{aligned} L : G &\to \mathrm{Perm}(X) \\ g &\mapsto L_g \end{aligned}$$

such that $\forall g_1, g_2 \in G$
$$L : g_1 g_2 \mapsto L_{g_1 g_2} = L_{g_1} \circ L_{g_2}.$$

If L exists, then X is called a *left group-space* (or *left G space*).

Remark 5.7 (Right action)**.** Obviously, it is also possible to define the right action, by requiring that
$$R : g_1 g_2 \mapsto R_{g_1 g_2} = R_{g_2} \circ R_{g_1}.$$

When they both exist, it is possible to show that the left and right actions are isomorphic.

[4] It is also a proper subgroup of $GL(n, \mathbb{R})$, because if $\det A = 1$ and $\det B = 1$ then $\det(AB) = 1 \cdot 1 = 1$.

[5] All the matrices in this group have $\det A = \pm 1$: if we restrict the set to have $\det A = +1$ we get the proper subgroup $SO(n)$ of the orthogonal group, and of the special linear transformations.

[6] This is because $|Au| = \sqrt{(Au)^\dagger (Au)} = \sqrt{u^\dagger A^\dagger A u} = \sqrt{u^\dagger u} = |u|$.

[7] Similarly to before, to find the dimension of this group: there are $2n^2$ entries, but $2n(n-1)/2$ constraints for the off-diagonal entries, and n for the diagonal entries (the imaginary part of the diagonal entries is zero because the matrix is guaranteed to be hermitian). Therefore $\dim U(n) = n^2$.

[8] Interestingly it has dimension $n(2n+1)$, and $\Omega \in Sp(2n, \mathbb{R})$.

Definition 5.8 (Orbit)**.** An orbit of an element $x \in X$, given a left action, is the set

$$O_x = \{x' \in X : \exists g \in G : x' = L_g(x)\}.$$

For example, if we take $X = \mathbb{R}^2$ and the group $G = SO(2)$: the orbit of a point x is the set

$$O_x = \left\{x' \in \mathbb{R}^2 : |x'| = |x|\right\}$$

because for each x' there exist a rotation (which is an element of G) that transforms x into x'.

In particular, belonging to the same orbit is an equivalence relation

1. $x \in O_x$ for all $x \in X$ (reflexive),
2. $x' \in O_x \iff x \in O_{x'}$ (symmetric),
3. $x' \in O_x, x'' \in O_{x'} \implies x'' \in O_x$ (transitive).

It is easy to show that $L : e \mapsto L_e = \mathrm{Id}$, thus $x \in O_x$. Then, if $x' \in O_x$ then $\exists g : x' = L_g(x)$, but if we consider $L_{g^{-1}}$ we find that

$$L_{g^{-1}} = (L_g)^{-1} : x' = L_g(x) \mapsto (L_g)^{-1} L_g(x) = x$$

and therefore $x \in O_{x'}$, which proves reflectivity. Transitivity can be proven in the same way, by applying this argument with the group product.

Definition 5.9 (Transitive left action)**.** Given a left action, it is said to be transitive if all elements of X belong to the same orbit (aka there is only one equivalence class).

For example, if we take $X = \mathbb{R}^2$ and $G = (\mathbb{R}^2, +)$ as the translations on the plane: the left action is transitive, because any point can be reached from every other point.

Definition 5.10 (Isotropy group, or "little group")**.** The isotropy group of $x \in X$ is defined as the subgroup of G whose elements (when L acts on them) leave x invariant

$$I_x = \{g \in G : L_g(x) = x\}.$$

Remark 5.11 (I_x is a subgroup)**.** In general, I_x is a subgroup because, for $g, g' \in I_x$ then

$$L_g(x) = L_{g'}(x) = x,$$

which implies that gg' is such that

$$L_{gg'}(x) = L_g(L_{g'}(x)) = L_g(x) = x$$

which means that $gg' \in I_x$.

Furthermore, $e \in I_x$ because $L_e(x) = E(x) = x$. And, finally, if $g \in I_x$ then

$$L_{g^{-1}}(x) = (L_g)^{-1}(x) = x$$

because $E(x) = (L_g)^{-1} L_g(x) = (L_g)^{-1}(x)$.

If the isotropy group I_x is equal to G, then x is called a *fixed point*: for example the origin in $\mathbb{R}^2$ is a fixed point for the $SO(2)$ rotations.

If we focus on the case when $X = G$, two special left actions are always defined:

1. "translation": $L : G \to \mathrm{Perm}(G)$ (an homomorphism) such that $L_g : x \mapsto g \cdot x$,[9]
2. "conjugation": such that $L_g : x \mapsto g \cdot x \cdot g^{-1}$.

[9] The nature of the definition is clear if we think about $G = \mathbb{R}^2$.